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Studierichting: Informatica
Oefeningenreeks 3A nr. 20.6

$$\begin{aligned}\frac{d}{dx} \left(\frac{3}{4(x^2 + 4x)^2} \right) &= \frac{3}{4} \frac{d}{dx} \left(\frac{1}{(x^2 + 4x)^2} \right) = \frac{3}{4} \left(\frac{(x^2 + 4x)^2 \cdot 0 - 1 \cdot 2(x^2 + 4x)(2x + 4)}{(x^2 + 4x)^4} \right) \\ &= \frac{3}{4} \left(-\frac{2(x^2 + 4x)(2x + 4)}{(x^2 + 4x)^4} \right) = -\frac{3(2x + 4)}{2(x^2 + 4x)^3} = -\frac{3(x + 2)}{(x^2 + 4x)^3}\end{aligned}$$

References